

DISTANCE EXPONENTS IN A RANDOM SERIES-PARALLEL GRAPH

ABSTRACT. Starting from one edge, replace every edge independently by two edges in series with probability p and by two parallel edges with probability $1 - p$. For every $p \in (1/2, 1)$, we characterize the exponential growth rate of the expected endpoint distance as the attained principal eigenvalue of an explicit nonlinear operator on decreasing quantile profiles. We also prove exact lower and upper Collatz–Wielandt formulas and a compatible invariant-measure variational formula. The critical identity at $p = 1/2$ is due to Chen, Derrida, Duquesne, and Shi, while the value at $p = 1$ is elementary; combined with those endpoint results, the interior characterization answers Benjamini’s Question 9.6 on the full parameter interval.

1. THE MODEL AND THE RESULT

Begin with one edge and two distinguished endpoints. At each generation, replace every current edge independently by two edges in series with probability p , or by two parallel edges with probability $q = 1 - p$. Let Δ_n be the distance between the distinguished endpoints after n generations, with $\Delta_0 = 1$. Root decomposition gives

$$(1) \quad \Delta_{n+1} \stackrel{d}{=} \begin{cases} \Delta_n^{(1)} + \Delta_n^{(2)}, & \text{with probability } p, \\ \min\{\Delta_n^{(1)}, \Delta_n^{(2)}\}, & \text{with probability } q, \end{cases}$$

where the two copies are independent. Put $m_n = \mathbb{E}\Delta_n$.

This model was studied by Hambly and Jordan [3]. Benjamini asked for the shape of the expected-distance exponent on $[1/2, 1]$ [1, Question 9.6]. Chen, Derrida, Duquesne, and Shi proved the critical identity $\delta(1/2) = 0$ and a sharp slightly-supercritical asymptotic [2]. The theorem proved here concerns only the residual interior interval $p \in (1/2, 1)$.

For a nonnegative integrable random variable X , write Q_X for its decreasing quantile on $(0, 1)$. For a nonnegative decreasing integrable profile Q , let U, V be independent uniform random variables on $(0, 1)$ and let B be an independent Bernoulli random variable with parameter p . Define

$$(2) \quad H_{p,Q} = B(Q(U) + Q(V)) + (1 - B)Q(\max\{U, V\})$$

and let $T_p Q$ be the decreasing quantile of $H_{p,Q}$. Since Q is decreasing, $Q(\max\{U, V\}) = \min\{Q(U), Q(V)\}$. If

$$M(Q) = \int_0^1 Q(u) du,$$

then $Q_n = T_p^n \mathbf{1}$ is exactly the decreasing quantile of Δ_n , and $m_n = M(Q_n)$.

For $p > 1/2$, define

$$(3) \quad K_p = \left\{ Q \geq 0 : Q \text{ is decreasing, } M(Q) = 1, \int_0^1 Q(u)^2 du \leq \frac{1}{2p-1} \right\}.$$

Theorem 1.1 (Interior spectral characterization). *For every $p \in (1/2, 1)$, the limit*

$$(4) \quad \rho(p) = \lim_{n \rightarrow \infty} m_n^{1/n} = \inf_{n \geq 1} m_n^{1/n}$$

exists, and $\delta(p) = \log \rho(p)$. There exists $Q_p \in K_p$ such that

$$(5) \quad T_p Q_p = \rho(p) Q_p.$$

The number $\rho(p)$ is the maximal eigenvalue among nonnegative, nonzero, integrable decreasing profiles, normalized to have mean one. Moreover,

$$(6) \quad \rho(p) = \max \{ \alpha \geq 0 : \exists Q \in K_p, T_p Q \geq \alpha Q \text{ a.e.} \},$$

$$(7) \quad \rho(p) = \inf \{ \beta > 0 : \exists Q \in K_p, \text{ess inf } Q > 0, T_p Q \leq \beta Q \text{ a.e.} \}.$$

The formulas are noncircular: the right-hand sides use only the explicit operator (2). No uniqueness of Q_p , elementary scalar formula for $\delta(p)$, or convergence of all normalized orbits is asserted.

2. SUBMULTIPLICATIVITY AND THE EXPONENTIAL RATE

The operator T_p is positively homogeneous and order preserving. It is also continuous in L^1 . Indeed, couple two profiles using the same B, U, V . The one-dimensional quantile representation of the Wasserstein distance gives

$$(8) \quad \|T_p Q - T_p R\|_1 = W_1(\mathcal{L}(H_{p,Q}), \mathcal{L}(H_{p,R})) \leq 2\|Q - R\|_1.$$

Fix a depth- n gate-labelled binary tree τ and let $\Phi_\tau(x_1, \dots, x_{2^n})$ be its endpoint-distance evaluation, using addition at a series gate and minimum at a parallel gate. The map Φ_τ is coordinatewise nondecreasing, positively homogeneous, and concave; these properties follow inductively because sums and pointwise minima preserve them. If $Y_1, \dots, Y_{2^n}$ are independent with mean one, Jensen's inequality yields

$$(9) \quad \mathbb{E}\Phi_\tau(Y_1, \dots, Y_{2^n}) \leq \Phi_\tau(1, \dots, 1).$$

Averaging over the gate labels shows that every mean-one profile Q satisfies

$$(10) \quad M(T_p^n Q) \leq m_n.$$

For depth- k leaf distances, put $Y_i = \Delta_k^{(i)}/m_k$, so $\mathbb{E}Y_i = 1$. Positive homogeneity and (9) give

$$\begin{aligned} m_{n+k} &= \mathbb{E}_\tau \mathbb{E} \Phi_\tau(\Delta_k^{(1)}, \dots, \Delta_k^{(2^n)}) \\ &= m_k \mathbb{E}_\tau \mathbb{E} \Phi_\tau(Y_1, \dots, Y_{2^n}) \leq m_k m_n. \end{aligned}$$

Thus $\log m_n$ is subadditive. The mean multiplier lies between $2p$ and $1+p$, so $(2p)^n \leq m_n \leq (1+p)^n$. Fekete's lemma proves (4); by the model's definition of the exponent, $\delta(p) = \log \rho(p)$.

3. A COMPACT INVARIANT PROFILE SET

For a decreasing profile define

$$S(Q) = \int_0^1 Q^2, \quad A(Q) = 2 \int_0^1 uQ(u) du, \quad C(Q) = 2 \int_0^1 uQ(u)^2 du.$$

When $M(Q) = 1$, the quantity $A(Q)$ is the mean of the minimum of two independent variables with quantile Q , so $0 \leq A(Q) \leq 1$. Directly from (2),

$$(11) \quad M(T_p Q) = d(Q) := 2p + qA(Q).$$

$$(12) \quad S(T_p Q) = 2pS(Q) + 2p + qC(Q).$$

The load-bearing anti-correlation inequality is

$$(13) \quad C(Q)M(Q) \leq A(Q)S(Q).$$

Indeed,

$$A(Q)S(Q) - C(Q)M(Q) = \int_0^1 \int_0^1 (u-v)Q(u)Q(v)(Q(v) - Q(u)) du dv \geq 0,$$

because Q is decreasing. For $M(Q) = 1$, equations (11)–(13) imply

$$(14) \quad S\left(\frac{T_p Q}{d(Q)}\right) \leq \frac{S(Q)}{d(Q)} + \frac{2p}{d(Q)^2} \leq \frac{S(Q) + 1}{2p}.$$

Since $d(Q) \geq 2p > 0$, every normalization denominator is nonzero. The bound $1/(2p-1)$ is invariant under (14); hence K_p is invariant under the mean-normalized map

$$N_p Q = \frac{T_p Q}{d(Q)}.$$

Starting from $\mathbf{1}$ also gives the finite-depth estimate

$$(15) \quad \frac{\mathbb{E}\Delta_n^2}{(\mathbb{E}\Delta_n)^2} \leq \frac{1}{2p-1}.$$

Lemma 3.1. *The set K_p is nonempty, convex, and compact in $L^1(0,1)$, and $N_p : K_p \rightarrow K_p$ is continuous.*

Proof. For a decreasing mean-one profile, $Q(u) \leq 1/u$ for $u > 0$. Helly selection on every $[\eta, 1]$, followed by a diagonal argument, gives an almost-everywhere convergent subsequence. The uniform L^2 bound implies uniform integrability, since

$$\int_E Q \leq |E|^{1/2} \left(\int_0^1 Q^2 \right)^{1/2} \leq \sqrt{\frac{|E|}{2p-1}}.$$

Vitali's theorem upgrades the convergence to L^1 . The mean is preserved, and Fatou's lemma preserves the L^2 bound. Continuity follows from (8), continuity of d , and $d(Q) \geq 2p$. $\square$

4. THE PRINCIPAL EIGENPROFILE

For $\varepsilon > 0$, set

$$T_{p,\varepsilon}Q = T_pQ + \varepsilon M(Q)\mathbf{1}, \quad N_{p,\varepsilon}Q = \frac{T_{p,\varepsilon}Q}{M(T_{p,\varepsilon}Q)}.$$

If a nonnegative random variable has mean m , second moment s , and is shifted by $c \geq 0$, then

$$(16) \quad \frac{s + 2cm + c^2}{(m + c)^2} \leq \frac{s}{m^2},$$

because the cross-multiplied difference is $c(2m+c)(s-m^2) \geq 0$. Consequently $N_{p,\varepsilon}$ is a continuous self-map of K_p . Schauder's fixed-point theorem gives $Q_\varepsilon \in K_p$ and $\lambda_\varepsilon > 0$ such that

$$(17) \quad T_{p,\varepsilon}Q_\varepsilon = \lambda_\varepsilon Q_\varepsilon, \quad M(Q_\varepsilon) = 1.$$

Moreover,

$$Q_\varepsilon \geq c_\varepsilon \mathbf{1}, \quad c_\varepsilon = \frac{\varepsilon}{\lambda_\varepsilon} > 0.$$

Order preservation and homogeneity imply

$$\lambda_\varepsilon^n Q_\varepsilon = T_{p,\varepsilon}^n Q_\varepsilon \geq c_\varepsilon T_p^n \mathbf{1}.$$

After integration and taking n -th roots, this yields $\lambda_\varepsilon \geq \rho(p)$.

By Lemma 3.1, along some $\varepsilon_j \downarrow 0$ one has $Q_{\varepsilon_j} \rightarrow Q_p$ in L^1 . Also

$$\lambda_\varepsilon = M(T_p Q_\varepsilon) + \varepsilon = d(Q_\varepsilon) + \varepsilon, \quad 2p + \varepsilon \leq \lambda_\varepsilon \leq 1 + p + \varepsilon,$$

so a subsequence converges to some λ . Passing to the limit in (17) gives $T_p Q_p = \lambda Q_p$ and $\lambda \geq \rho(p)$. Conversely, iteration and (10) give

$$\lambda^n = M(T_p^n Q_p) \leq m_n,$$

so $\lambda \leq \rho(p)$. Thus $\lambda = \rho(p)$, proving (5). The argument does not assume convergence of the actual orbit $N_p^n \mathbf{1}$.

If an admissible normalized eigenprofile satisfies $T_p Q = \lambda Q$, then (10) gives $\lambda^n \leq m_n$ and hence $\lambda \leq \rho(p)$. This proves maximality of the numerical eigenvalue, not uniqueness of its eigenprofile.

Suppose next that $Q \in K_p$ and $T_p Q \geq \alpha Q$. Iteration and (10) give $\alpha^n \leq m_n$, hence $\alpha \leq \rho(p)$; equality is attained by Q_p . This proves (6).

For the upper formula, assume $Q \in K_p$, $Q \geq c\mathbf{1}$ almost everywhere for some $c > 0$, and $T_p Q \leq \beta Q$. Then

$$cT_p^n \mathbf{1} \leq T_p^n Q \leq \beta^n Q,$$

so $cm_n \leq \beta^n$ and $\rho(p) \leq \beta$. Conversely, the perturbed fixed points satisfy

$$T_p Q_\varepsilon = \lambda_\varepsilon Q_\varepsilon - \varepsilon \mathbf{1} \leq \lambda_\varepsilon Q_\varepsilon,$$

with $\text{ess inf } Q_\varepsilon > 0$, and every cluster value of λ_ε is $\rho(p)$. This proves (7) and completes the proof of Theorem 1.1.

5. AN INVARIANT-MEASURE FORMULA

The spectral characterization has a compatible dynamical form. Let $\mathcal{I}_p$ be the Borel probability measures on K_p invariant under N_p , and put $f_p(Q) = \log d(Q)$.

Proposition 5.1. *For every $p \in (1/2, 1)$,*

$$(18) \quad \delta(p) = \max_{\Pi \in \mathcal{I}_p} \int_{K_p} f_p(Q) d\Pi(Q).$$

Proof. For $Q_j = N_p^j \mathbf{1}$,

$$\log m_N = \sum_{j=0}^{N-1} f_p(Q_j).$$

Every weak limit of the empirical occupation measures is invariant and has average $\delta(p)$. Conversely, homogeneity gives, for every mean-one Q ,

$$\log M(T_p^n Q) = \sum_{j=0}^{n-1} f_p(N_p^j Q).$$

Integrating against an invariant Π and applying (10) yields

$$n \int f_p d\Pi \leq \log m_n.$$

Taking the infimum over n proves the reverse inequality. The Dirac mass at Q_p attains the maximum, but the proof does not assert that every maximizing invariant measure is a fixed-point mass. $\square$

6. ENDPOINTS AND EXACT SCOPE

Theorem 1.1 addresses only $p \in (1/2, 1)$. At the critical endpoint, Chen, Derrida, Duquesne, and Shi proved $\delta(1/2) = 0$ [2]; the compactness argument here does not extend to $p = 1/2$ because the bound $1/(2p - 1)$ diverges. At $p = 1$, the distance is deterministically 2^n , so $\delta(1) = \log 2$. Thus the interior theorem, the prior critical theorem, and the elementary right endpoint together characterize the full interval in the sense asked by Benjamini's Question 9.6.

The characterization is exact but not an elementary scalar formula in p . It does not prove uniqueness of Q_p , global convergence of normalized laws, monotonicity or convexity of δ , or any compactness statement at the critical endpoint.

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